

## **MINI-PROJECT 5.3**

# **DETECTING THE ANOMALOUS ACTIVITY OF A SHIP'S ENGINE.**

Stakeholders Report

by

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## Introduction

To optimize fleet performance, the company needs a high-performance anomaly detection system to monitor engine health and pinpoint specific units requiring intervention. By facilitating proactive and targeted maintenance, this system will enhance operational safety and efficiency. These improvements are expected to drive significant downstream benefits, including heightened customer satisfaction and increased corporate profitability.

## Methodology

The ship's engine data set has six features with 19,535 observations. Three methods of anomaly detection approaches were adopted for the processes as highlighted on the table 1 below. The data was scaled to perform OCSVM and PCA using z-score standardization scaling method.

**Table 1:** *Anomaly Detection Methods*

| Method           | Type        | Operation                                                                                                                           | Key parameters                                   |
|------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| IQR              | Statistical | Anomalous points fall above the upper bound limit ( $Q3 + 1.5 \cdot IQR$ ) and below the lower bound limit ( $Q1 - 1.5 \cdot IQR$ ) | Number of features that contain anomalous points |
| One-Class SVM    | ML          | Outliers were detected by separating them from normal data with a hyperplane in high-dimensional space                              | Nu, gamma                                        |
| Isolation Forest | ML          | Ensemble method that isolates outliers using random splits; shorter path length indicates outliers                                  | Contamination, n-estimates                       |

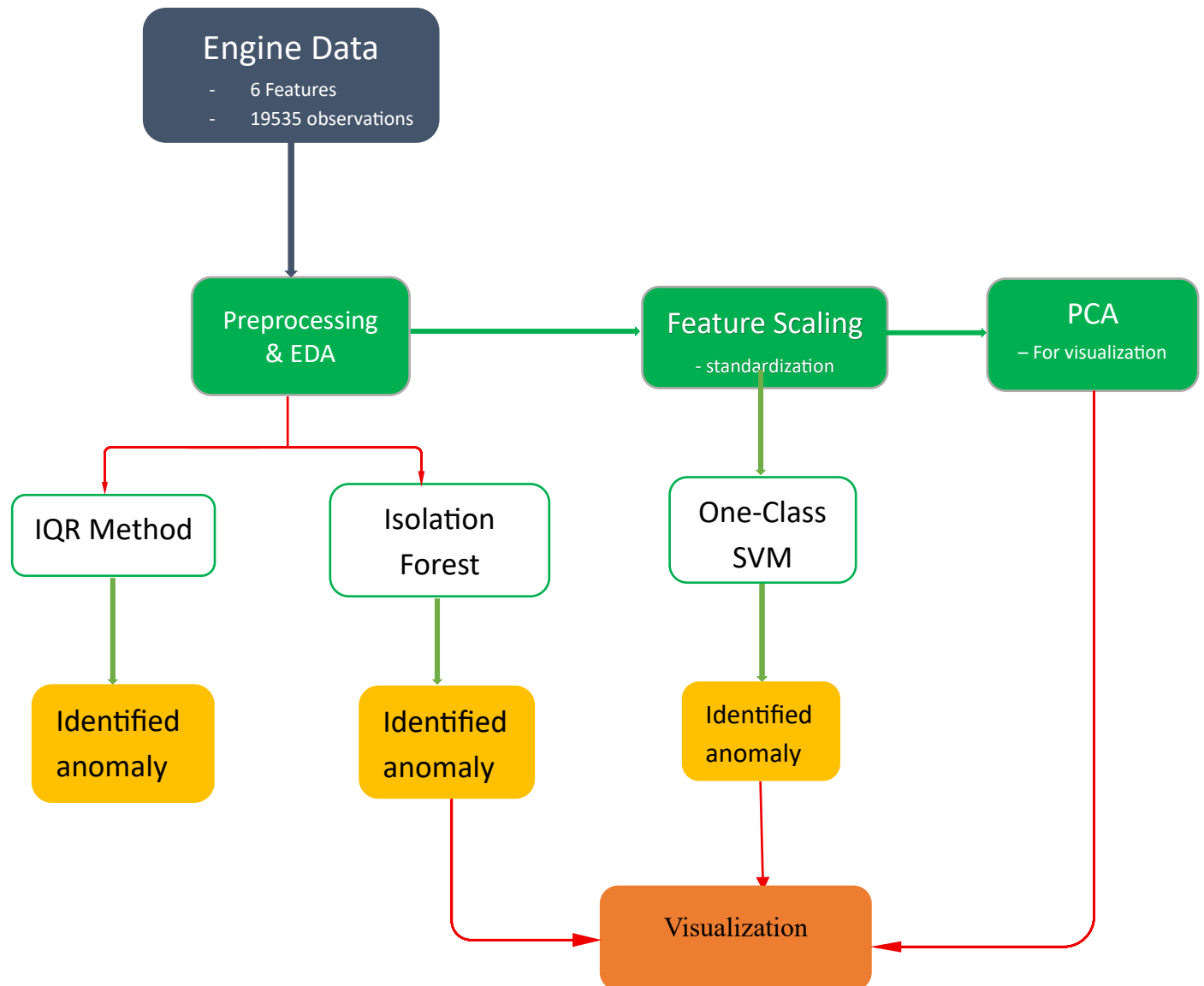
**Abbreviations:** IQR=Interquartile Range; ML= Machine Learning; SVM = Support Vector Machine

Presented in fig. 1 is the summary of the approach in executing the project. Data processing included an assessment of missing and duplicate values, computing descriptive statistics and visualizing the distribution and extreme values for each feature using box plots.

The Interquartile range (IQR) method was initially used to detect anomalies within each feature independently. This univariate analysis was then extended into a multivariate evaluation by determining the total number of observations classified as outliers based on how many individual features simultaneously flagged an anomaly.

The One-Class SVM and Isolation Forest models underwent hyperparameter tuning; Table 1 represents the different values tested during this optimization process. To facilitate a two-

dimensional (2D) visualization of the One-Class SVM and Isolation Forest models outputs, Principal Component Analysis (PCA) was employed as a dimensionality reduction technique.



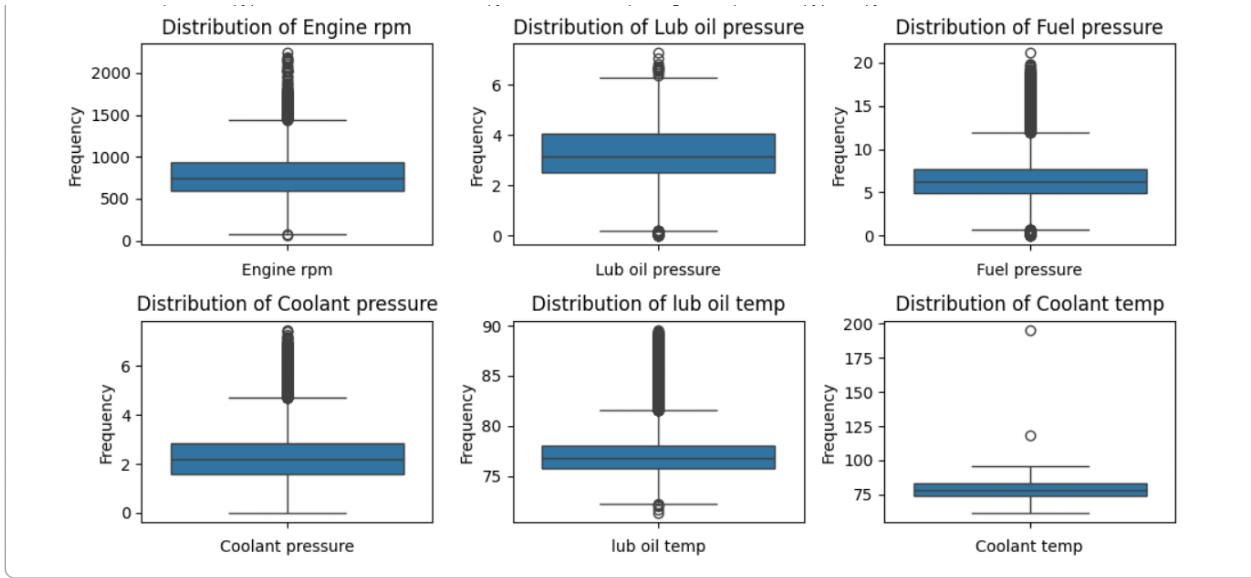
**Figure 1:** *Schematic Overview of Engine Data Processing and Anomaly Identification Pipeline* (Source: Author)

**Abbreviations:** EDA – Exploratory Data Analysis; IQR – Interquartile Range; ML – Machine Learning; PCA – Principal Component Analysis; SVM – Support Vector Machine.

Given the sensitivity of both PCA and One-Class SVM to variable magnitudes, feature scaling was implemented via Z-score standardization to prevent feature dominance. The anomalies identified by each of the three methods were compared to establish a measure of reliability. This process

highlighted the overlap between models, ensuring the robust identification of true outliers through methodological agreement.

Results and interpretation



1. Box plot Visualization

Figure 2: Box plot showing the distribution of anomalies in the features (Source: Author)

Analysis of the boxplots indicates that outliers are present across all features. While every variable shows some level of deviation, **Engine RPM, Lub Oil Pressure, Fuel Pressure, and Coolant Temperature** display the most significant or frequent anomalies. These areas represent primary targets for an anomaly detection system, with the **Coolant Temperature** specifically showing extreme maximums that suggest critical operational concerns as shown in figure 2.

2. IQR Method.

Table 2: Distribution of Anomalous Observations Across Multiple Features

| Analysis type | Threshold condition | Observations | Percentage | Finding                   |
|---------------|---------------------|--------------|------------|---------------------------|
| Univariate    | $\geq 1$            | 4,636        | 23.7%      | Potential False Positives |
| Multivariate  | $\geq 2$            | 422          | 2.2%       | Significant Anomalies     |

|              |          |    |      |                    |
|--------------|----------|----|------|--------------------|
| Multivariate | $\geq 3$ | 11 | 0.1% | Critical Anomalies |
| Multivariate | $\geq 4$ | 0  | 00   | No anomaly         |

In IQR algorithm, 4,636 number of observations representing up to 23.7% had an anomaly in at least one feature, making univariate analysis inappropriate for anomaly detection. However, multivariate analysis show that 422 observations representing about 2.2% had anomaly in a least ( $\geq 2$ ) features. The threshold was increased to anomalies in at least three ( $\geq 3$ ) features resulting in only 11 observations which represent 0.1%. No observation was highlighted with anomalies in at least four ( $\geq 4$ ) features. This therefore indicated that in the IQR method, observations were flagged as outliers when anomalies occur in at least ( $\geq 2$ ) features.

### 3. One-Class SVM Method.

In One-Class SVM, the number of anomalies vary with the tuning of the parameters as outlined on the table 3.

**Table 3:** *Anomaly values with varied parameters.*

| Gamma | Nu   | Number of anomalies | Percentage |
|-------|------|---------------------|------------|
| 0.1   | 0.01 | 189                 | 1.0%       |
| 0.5   | 0.01 | 611                 | 3.1%       |
| 0.1   | 0.05 | 978                 | 5.0%       |

The One-Class SVM model is highly sensitive to the nu parameter than the gamma, and it serves as an effective control for setting the desired proportion of anomalies.

### 4. Isolation Forest Method.

In Isolation Forest, adjusting the factor allowed the number of anomalies to be detected. Any value of contamination up to 0.05 was appropriate.

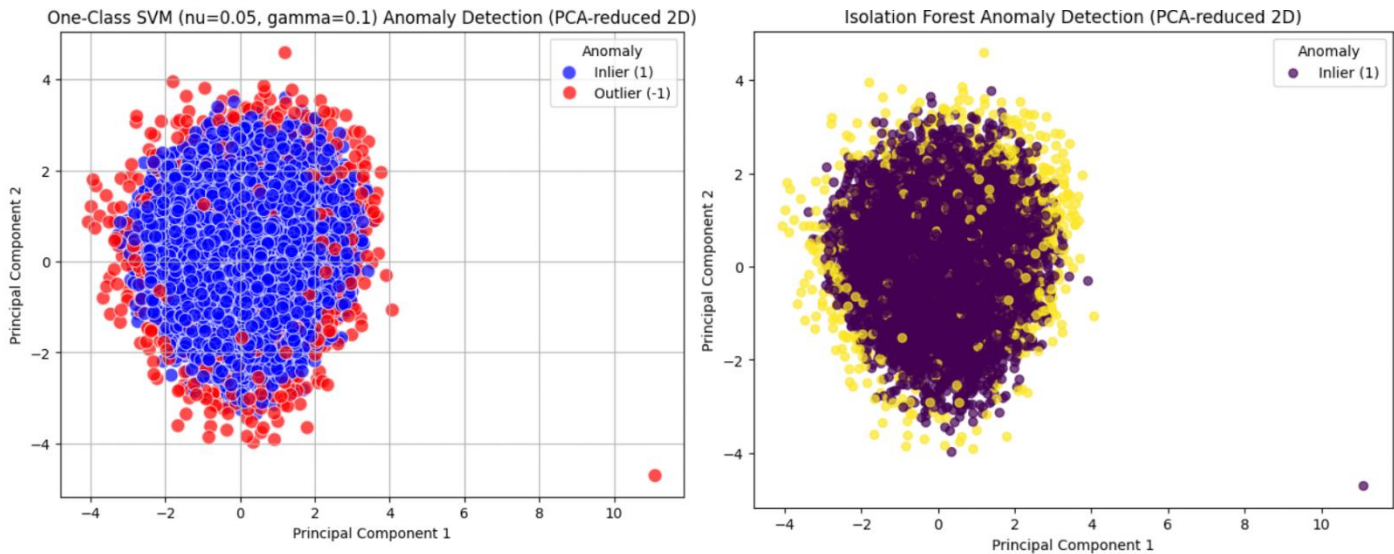
**Table 4:** *Distribution of anomalous observations across varied parameters.*

| Contamination | n_estimators | Number of outliers | Percentage |
|---------------|--------------|--------------------|------------|
| <b>0.01</b>   | 100          | 196                | 1.0%       |
| <b>0.03</b>   | 50           | 587                | 3.0%       |
| <b>0.05</b>   | 100          | 977                | 5.0%       |

Visual representations of the One-class SVM and Isolation Forest results—set at a 5% outlier threshold—reveal that anomalies are concentrated along the cluster's periphery, as shown in the 2-feature PCA plot (Figure 3).

### 5. PCA Method:

There is no clustering of normal and outlier datapoints from the resulting PCA plots (figure 3). However, these results require deeper analysis to be operationally useful. Since PCA dimensions do not represent original variables, we cannot currently determine which specific engine features are driving these anomalous readings.



**Fig. 3:** Visual representation of the One-Class SVM and Isolation Forest models (source: Author)

When results of each method were compared, using the 5% anomalies for One-Class SVM and Isolation Forest, 225 (1.2%), observations were classified as an anomaly by all three methods. This highlights a moderate degree of overlap but emphasizes the need for careful consideration when selecting an appropriate method, as they may classify different observation differently

### Conclusion and recommendation

Observations where at least **two features** exhibit anomalies need to be prioritized to minimize false positives and focus resources on the 422 most significant cases identified. Also, utilizing a "consensus model" approach. Observations flagged by both One-Class SVM and Isolation Forest (the 225 core anomalies) should be the primary focus for the initial monitoring shortlist.

A follow-up diagnostic phase should be conducted to map abstract PCA components back to physical engine features (e.g., Coolant Temp, RPM) to provide maintenance crews with actionable data. And also, establishing a feedback loop where maintenance findings are used to retrain the models, further improving the precision of the 5% outlier threshold.